

Asymptotic properties of MLE

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Param Inf.

$$\mathcal{L}(\theta|x) = f(x|\theta) \stackrel{\text{if } x_i \text{ iid}}{=} \prod_{i=1}^n f(x_i|\theta)$$

Why do we maximise $\mathcal{L}(\theta|x)$? Does it guarantee that $\hat{\theta}$ = the MLE represents the true value of θ ?

Suppose $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \underbrace{f(x|\theta)}_{\text{identifiable}}; \theta \in \Theta$

[if θ_0 = true value of θ
 $\Leftrightarrow X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f(x|\theta_0)$]

$$\hat{\theta} = \underset{\theta \in \Theta}{\operatorname{argmax}} \mathcal{L}(\theta|x)$$

Support $\mathcal{X} = \{x: f(x|\theta) > 0\}$ is free of θ

$$\mathbb{P}_{\theta_0}(\mathcal{L}(\theta_0|x) \geq \mathcal{L}(\theta|x)) \rightarrow 1 \text{ as } n \rightarrow \infty$$

$$\frac{\prod_{i=1}^n f(x_i|\theta)}{\prod_{i=1}^n f(x_i|\theta_0)}$$

$$\mathbb{P}_{\theta_0} \left(\prod_{i=1}^n \frac{f(x_i|\theta_0)}{f(x_i|\theta)} > 1 \right)$$

$$= \mathbb{P}_{\theta_0} \left(\frac{1}{n} \sum \underbrace{\ln \left(\frac{f(x_i|\theta_0)}{f(x_i|\theta)} \right)}_{Z_i} > 0 \right)$$

$$= 1 - \mathbb{P}_{\bar{Z}_n}(0)$$

$$\rightarrow 1 - F_c(0) = 1$$

$$[\because F_c(x) = \mathbb{1}_{\{x > c\}}]$$

$$\begin{aligned} \mathbb{E}_{\theta_0}[Z_i] &= -\mathbb{E}_{\theta_0} \left[\ln \frac{f(x_i|\theta)}{f(x_i|\theta_0)} \right] \end{aligned}$$

$$> -\ln \underbrace{\mathbb{E}_{\theta_0} \left[\frac{f(x_i|\theta)}{f(x_i|\theta_0)} \right]}_{=1}$$

$$= 0$$

$$\frac{1}{n} \sum_{i=1}^n Z_i \xrightarrow{P} \mathbb{E}_{\theta_0} Z_i > 0$$

$$\Rightarrow \bar{Z}_n \xrightarrow{d} c > 0$$

An estimator δ_n of θ is called "Consistent and Asymptotically Normal" (CAN) estimator if

$$1) \delta_n \xrightarrow{P} \theta$$

$$2) \text{ For some real sequence } \{a_n\} \\ a_n(\delta_n - \theta) \xrightarrow{d} N(0, \sigma^2(\theta))$$

Notation: $\delta_n \sim \underbrace{AN(\theta, \underbrace{\frac{\sigma^2(\theta)}{a_n^2}}_{\text{asymptotically normal}})}_{\text{asymptotically normal}} \longrightarrow \text{Asymptotic variance of } \delta_n \equiv AV(\delta_n) = \frac{\sigma^2(\theta)}{a_n^2}$

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poi}(\lambda)$. Find a CAN est. of $\psi(\lambda) = e^{-\lambda}$

$$\text{UMVUE} \longrightarrow \left(1 - \frac{1}{n}\right)^{\sum X_i}$$

MLE of $e^{-\lambda} = e^{-\bar{X}}$. Is it CAN

$$\sqrt{n}(\bar{X} - \lambda) \xrightarrow{d} N(0, \lambda)$$

$$\Rightarrow \sqrt{n}(e^{-\bar{X}} - e^{-\lambda}) \xrightarrow{d} N(0, \lambda e^{-2\lambda})$$

$\therefore e^{-\bar{X}}$ is a CAN estimator with $AV = \frac{\lambda}{n} e^{-2\lambda} \rightarrow \text{CRLB? (Yes)}$

MLE in one-parameter exponential family (CAN)

$$(\sqrt{n}(\hat{\theta}_{MLE} - \theta) \xrightarrow{d} N(0, \frac{1}{I(\theta)}))$$

Let $X_1, \dots, X_n \stackrel{iid}{\sim} f(x|\theta) := h(x)e^{\eta(\theta)T(x) - B(\theta)} \quad \forall \theta \in \Theta$

Find $I_{X_1}(\theta)$: $= e^{\eta(\theta)T(x) - B(\theta) + s(x)}$ where $s(x) = \ln(h(x))$

$$\ln f(x|\theta) = \eta(\theta)T(x) - B(\theta) + s(x)$$

$$\frac{\partial}{\partial \theta} \ln f(x|\theta) = \eta'(\theta)T(x) - B'(\theta)$$

$$\mathbb{E}\left[\frac{\partial}{\partial \theta} \ln f(x|\theta)\right] = \mathbb{E}\left[\eta'(\theta) T(x) - B'(\theta)\right]$$

$$\Rightarrow \mathbb{E} T(x) = \frac{B'(\theta)}{\eta'(\theta)} = \psi(\theta) \text{ say}$$

$$\frac{\partial^2}{\partial \theta^2} \ln f(x|\theta) = \eta''(\theta) T(x) - B''(\theta)$$

$$\begin{aligned} \therefore I_{X_1}(\theta) &= -\mathbb{E}\left[\frac{\partial^2}{\partial \theta^2} \ln f(x|\theta)\right] \\ &= B''(\theta) - \eta''(\theta) \mathbb{E} T(x) \\ &= \frac{B''(\theta) \eta'(\theta) - \eta''(\theta) B'(\theta)}{\eta'(\theta)} \end{aligned}$$

$$\text{MLE: } \mathcal{L}(\theta|x) = e^{\eta(\theta) \sum_{i=1}^n T(x_i) - n B(\theta) + \sum_{i=1}^n S(x_i)}$$

$$\frac{\partial}{\partial \theta} \ln \mathcal{L}(\theta|x) = \eta'(\theta) \sum_{i=1}^n T(x_i) - n B'(\theta) = 0$$

$$\frac{1}{n} \sum_{i=1}^n T(x_i) = \frac{B'(\theta)}{\eta'(\theta)} = \psi(\theta) \dots (*)$$

if $\hat{\theta}$ is the solⁿ of $(*)$ then

$$\frac{\partial^2}{\partial \theta^2} \ln \mathcal{L}(\theta|x) = \eta''(\theta) \sum_{i=1}^n T(x_i) - n B''(\theta) = 0 \quad \sum_{i=1}^n T(x_i) = n \psi(\hat{\theta})$$

$$\begin{aligned} \frac{\partial^2}{\partial \theta^2} \ln \mathcal{L}(\theta|x) \Big|_{\hat{\theta}} &= \eta''(\hat{\theta}) \underbrace{n \psi(\hat{\theta})}_{\frac{B'(\hat{\theta})}{\eta'(\hat{\theta})}} - n B''(\hat{\theta}) \\ &= -n \left[B''(\hat{\theta}) - \frac{\eta''(\hat{\theta}) B'(\hat{\theta})}{\eta'(\hat{\theta})} \right] = -n I_{X_1}(\hat{\theta}) < 0 \end{aligned}$$

$$\delta_n := \frac{1}{n} \sum_{i=1}^n T(X_i) \xrightarrow{P} \psi(\theta)$$

$$\text{Assume } \psi^{-1} \text{ exist, } \Rightarrow \psi^{-1}(\delta_n) = \hat{\theta}_{MLE} \xrightarrow{P} \theta$$

(By continuous mapping theorem)

$$\text{We have } \delta_n := \frac{1}{n} \sum_{i=1}^n T(X_i) \sim AN\left(\psi(\theta), \frac{\sigma^2(\theta)}{n}\right) \text{ (by CLT)}$$

$$\text{where } \sigma^2(\theta) = V(T(X))$$

$$V\left(\frac{\partial}{\partial \theta} \ln f(X|\theta)\right) = (\eta'(\theta))^2 V(T(X))$$

$$V(T(X)) = \frac{I_{X_1}(\theta)}{(\eta'(\theta))^2}$$

Assume ψ^{-1} exists, continuously differentiable

$$\sqrt{n}(\delta_n - \psi(\theta)) \xrightarrow{d} N(0, \sigma^2(\theta))$$

$$\Rightarrow \sqrt{n}(\underbrace{\psi^{-1}(\delta_n)}_{\hat{\theta}_{MLE}} - \theta) \xrightarrow{d} N(0, \sigma^2(\theta) \left(\frac{\partial}{\partial \theta} \psi^{-1}(\theta)\right)^2)$$

$$\begin{aligned} \text{Show } \frac{\partial}{\partial \theta} \psi^{-1}(\theta) &= \frac{1}{\frac{\partial \psi}{\partial \theta}} \text{ (HW)} \\ &= \frac{\eta'(\theta)}{I_{X_1}(\theta)} \end{aligned}$$

$$\begin{aligned} \psi'(\theta) &= \frac{E[\eta' - \theta' \eta'']}{(\eta')^2} \\ &= \frac{I_{X_1}(\theta)}{\eta'(\theta)} \end{aligned}$$

$$AV(\hat{\theta}) = \sigma^2(\theta) \left(\frac{\partial}{\partial \theta} \psi^{-1}(\theta)\right)^2 = \frac{I_{X_1}(\theta)}{\eta'(\theta)^2} \left(\frac{\eta'(\theta)}{I_{X_1}(\theta)}\right)^2 = \frac{1}{I_{X_1}(\theta)}$$

HW If $X_1, \dots, X_n \stackrel{iid}{\sim}$ with mean $\mu(\theta)$ and var $\sigma^2(\theta)$ and $\frac{d\mu(\theta)}{d\theta}$ is cont. then

$$\sqrt{n}(\bar{\mu}^{-1}(\bar{X}) - \theta) \xrightarrow{d} N\left(0, \frac{\sigma^2(\theta)}{\left(\frac{d\mu}{d\theta}\right)^2}\right)$$

Consistency of MLE (Lehmann - Large Sample Theory)

Suppose $X_1, \dots, X_n \stackrel{iid}{\sim} f(x|\theta)$, $\theta \in \Theta \subseteq \mathbb{R}$ satisfying the conditions

- (C1) $f(x|\theta_1) \neq f(x|\theta_2) \forall \theta_1 \neq \theta_2, \theta_1, \theta_2 \in \Theta$
- (C2) Support $\chi = \{x: f(x|\theta) > 0\}$ is free of θ
- (C3) Θ is an open interval
- (C4) $\frac{\partial}{\partial \theta} f(x|\theta)$ exists
- (C5) If $f(x|\theta)$ is pdf (pmf), then it is cont. (discrete) fn. of x , $\forall x \in \chi$

Then $\exists$ a sequence $\hat{\theta}_n$ of local maxima of the likelihood fn. $L(\theta|x)$ such that

$$\hat{\theta}_n \xrightarrow{P} \theta \quad (\forall \theta \in \Theta)$$

If $\hat{\theta}_n$ is the unique root of the likelihood eqn

$$\frac{\partial}{\partial \theta} \ln L(\theta|x) = 0, \text{ then } \underset{\text{MLE}}{\hat{\theta}_n} \xrightarrow{P} \theta$$

Ex: $X_1, \dots, X_n \stackrel{iid}{\sim}$ Logistic $f(x|\theta) = \frac{e^{-(x-\theta)}}{\{1 + e^{-(x-\theta)}\}^2}, \theta \in \mathbb{R}$

Show: Likelihood eqnⁿ has $\hat{\theta}$ root which is the MLE

HW and $\hat{\theta} \xrightarrow{P} \theta$

HW $I_{X_1}(\theta) = \frac{1}{3}$ Hence $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, 3)$